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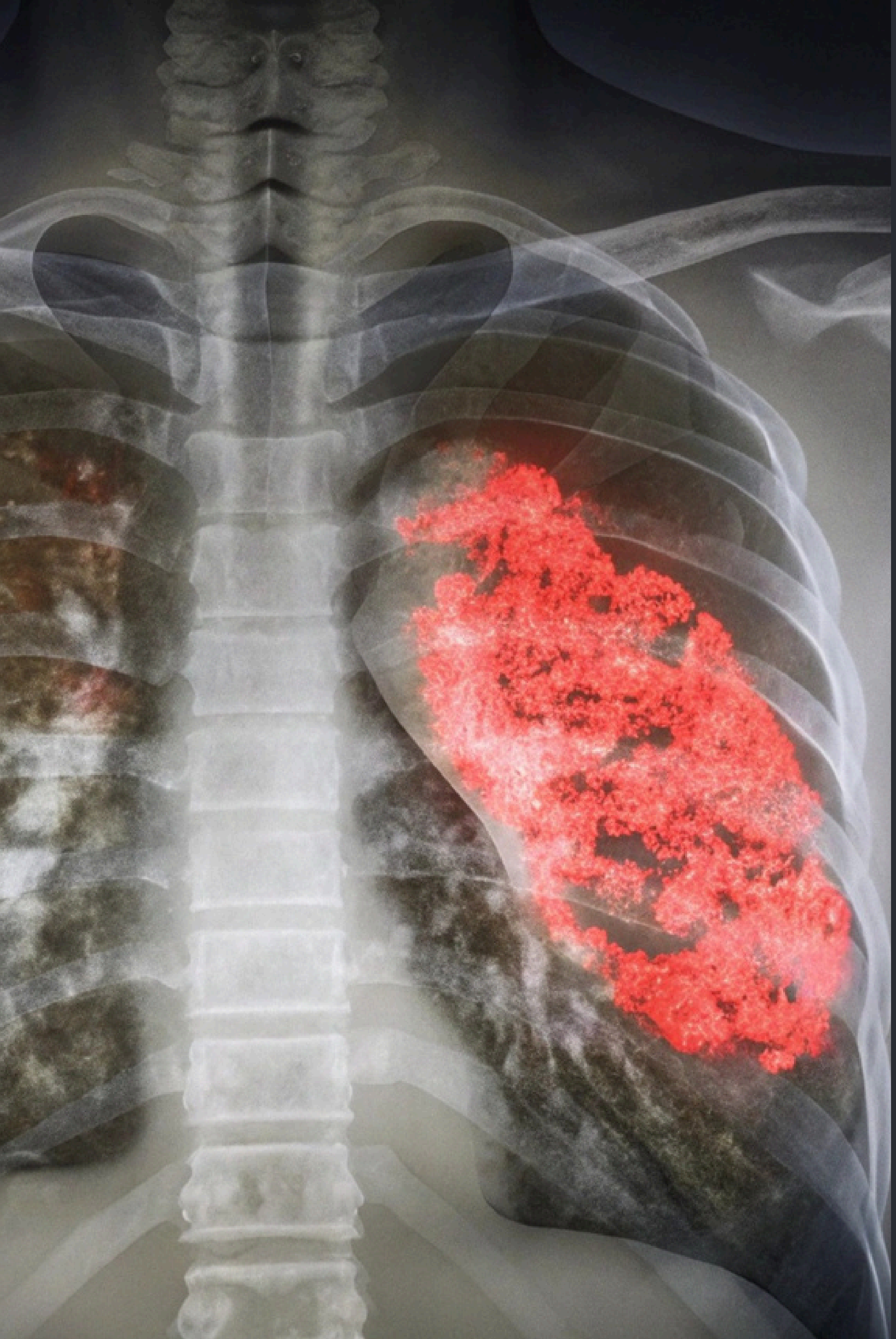

Pneumonia Detection Using Optimized Deep Learning Technique

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New Mansoura University. Computer Science and Engineering IC-FTAI 2025 | Alexandria, Egypt

The Clinical Problem

Global Disease Burden

Pneumonia is a form of **acute respiratory infection** that is most commonly **caused by viruses or bacteria**. It can cause mild to life-threatening illness in people of all ages, however it is the **single largest infectious cause of death in children worldwide**. Pneumonia **killed** more than **808 000** children under the age of 5 in 2017, accounting for **15% of all deaths of children under 5 years**. People at-risk for pneumonia also include adults over the **age of 65** and **people with preexisting health problems**.

-World Health Organization

What happens during infection is that alveoli fill with fluid which impairs gas exchange

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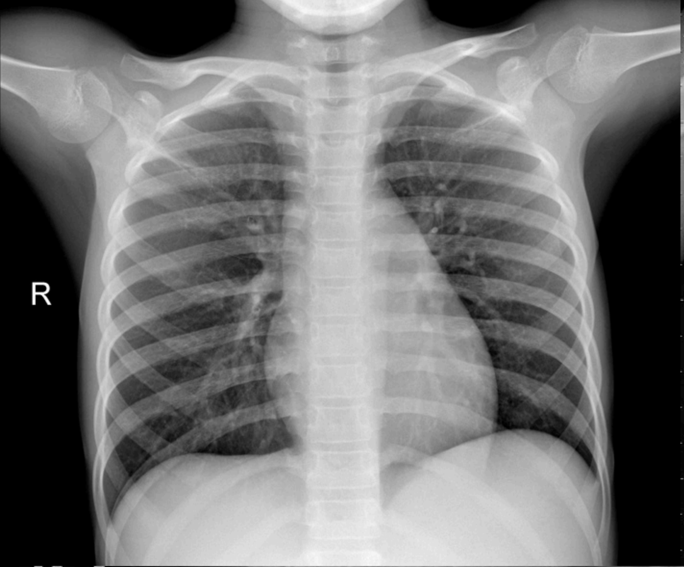

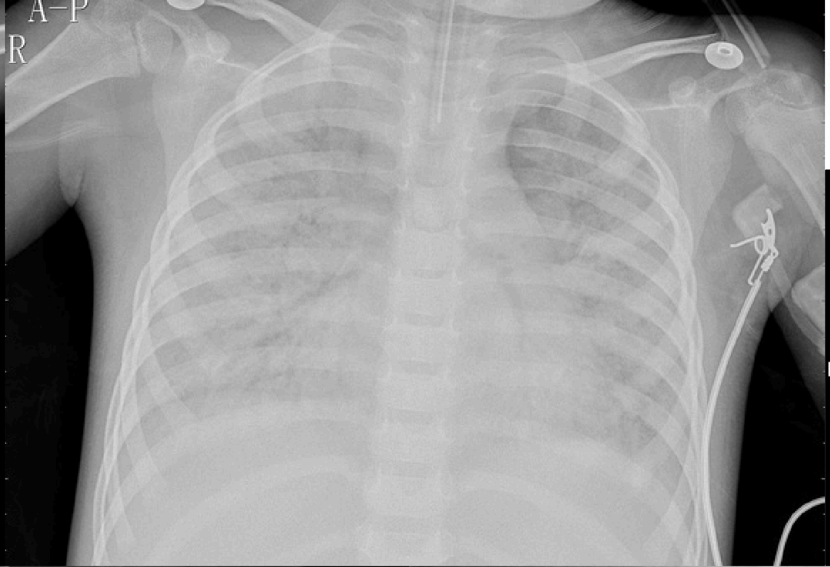

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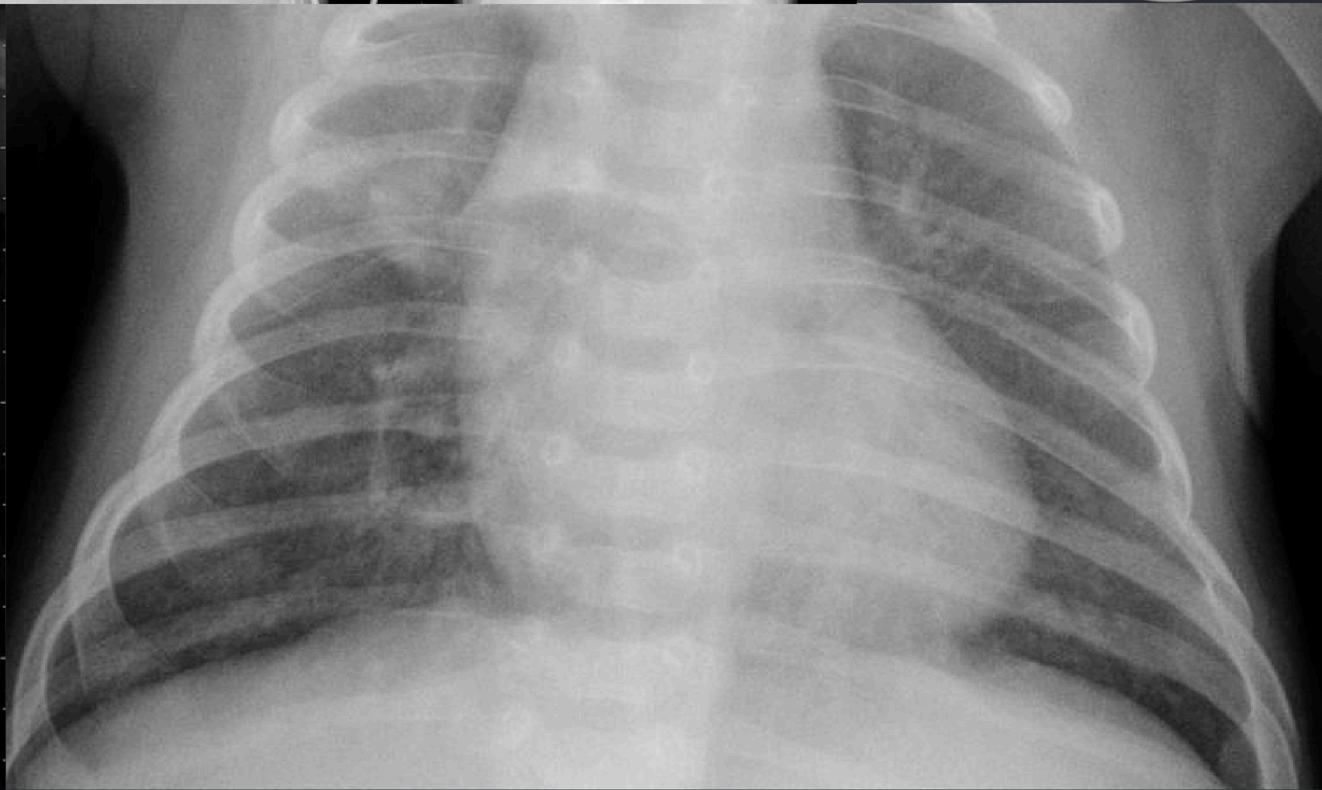
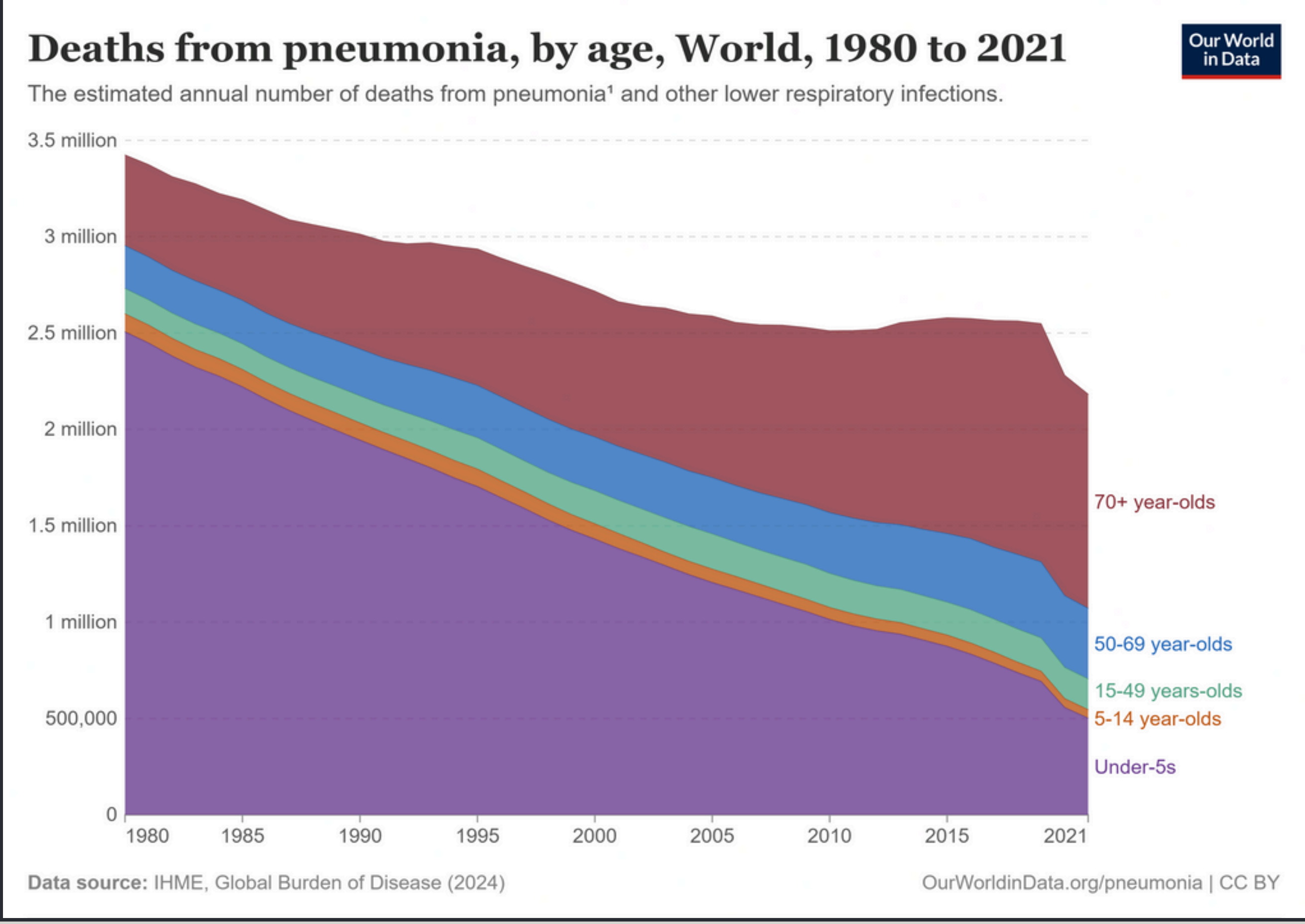


It was the cause of millions of global deaths anually

and it was mostly due to:

- Delays in diagnosis due to Doctor unavailability
- Missclassifications of sick patients
- misunderstanding problem complexity



Diagnostic Delays

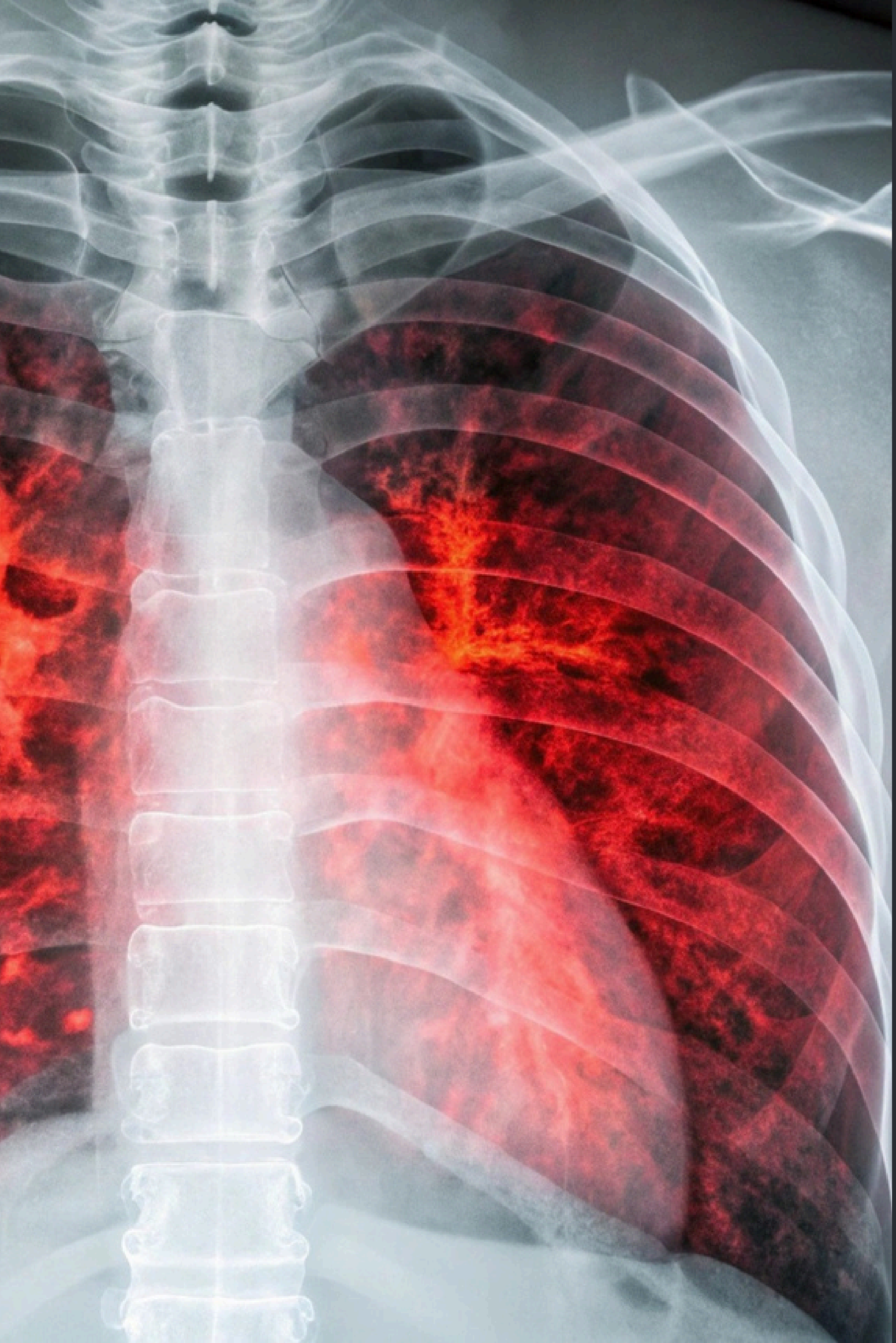
Manual radiological review creates bottlenecks, delaying treatment initiation and patient care decisions.

CNN Overfitting

Deep neural networks often overfit to training data, reducing generalization and clinical reliability in real-world deployments.

Hyperparameter Optimization

Traditional optimization methods fail to efficiently navigate high-dimensional parameter spaces inherent to modern deep learning architectures.



Research Objective

Goal

Make pneumonia detection More Reliable in real-world application using Genetic Algorithm-optimized VGG16 convolutional neural networks.

Challenge

Overcome overfitting in deep learning models and optimize weights for clinical deployment.

Impact

Enable rapid, accurate pneumonia diagnosis to support early intervention, **Minimize Human mistakes** and **Reduce diagnostic delays** in resource-constrained settings.

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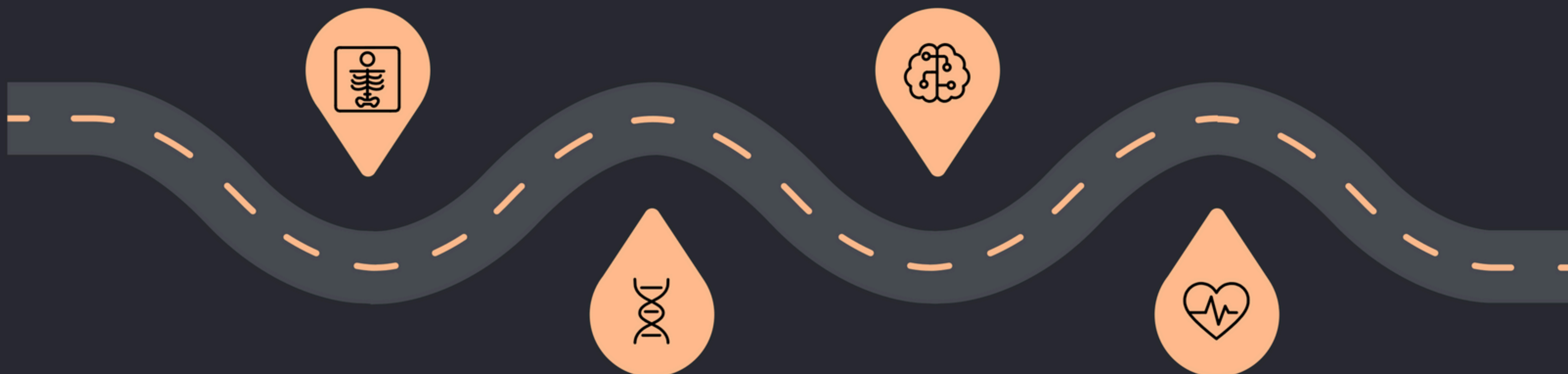
Methodology



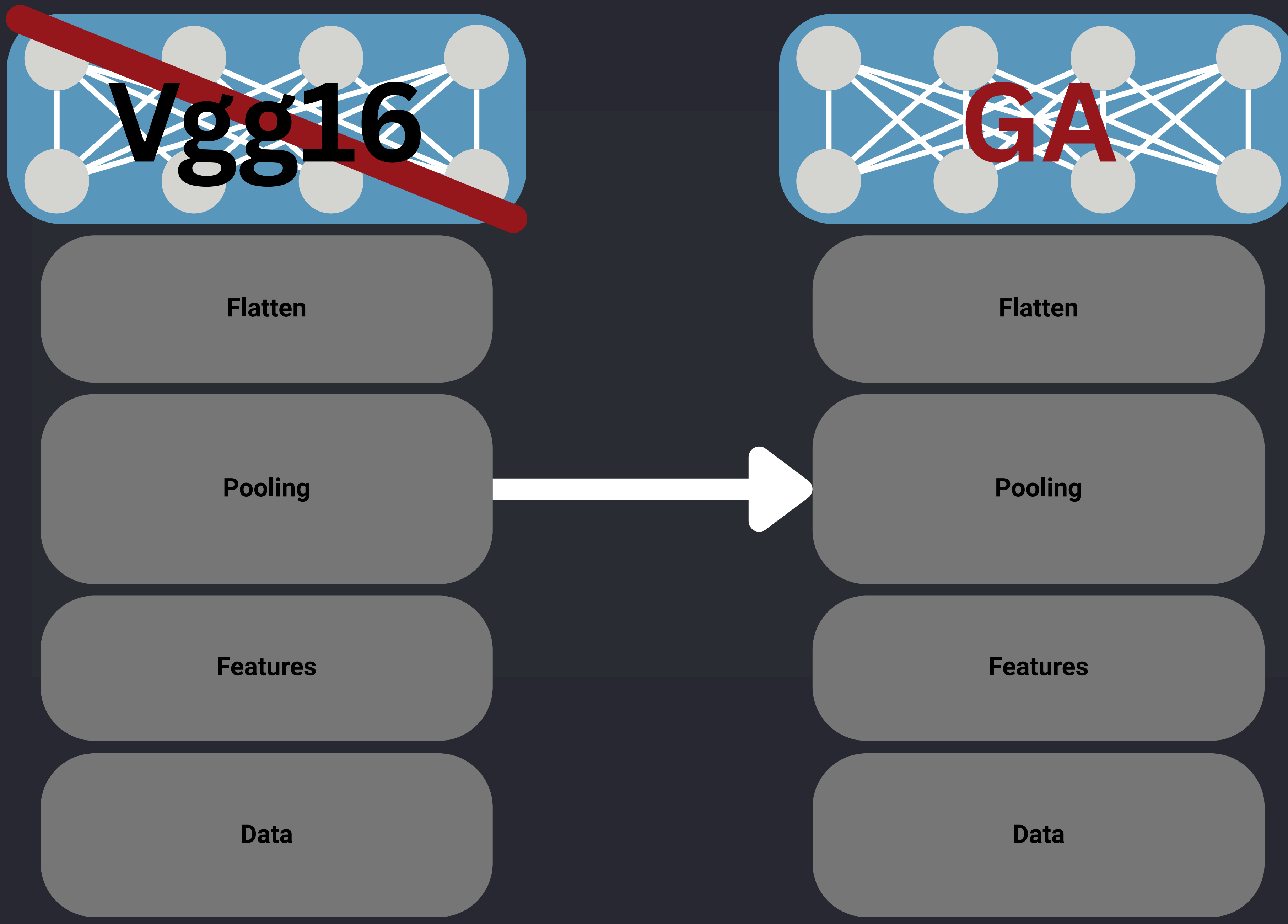
Input Images
Chest X-ray dataset

GA Optimization
Selection, crossover, + mutation

Cloning model
architecture and fitting

VGG16 Model
Choosing Best Parent and Looping until weight finalization

Detection Output
Classified pneumonia result


Vgg16
Flatten
Pooling
Features
Data
GA
Flatten
Pooling
Features
Data

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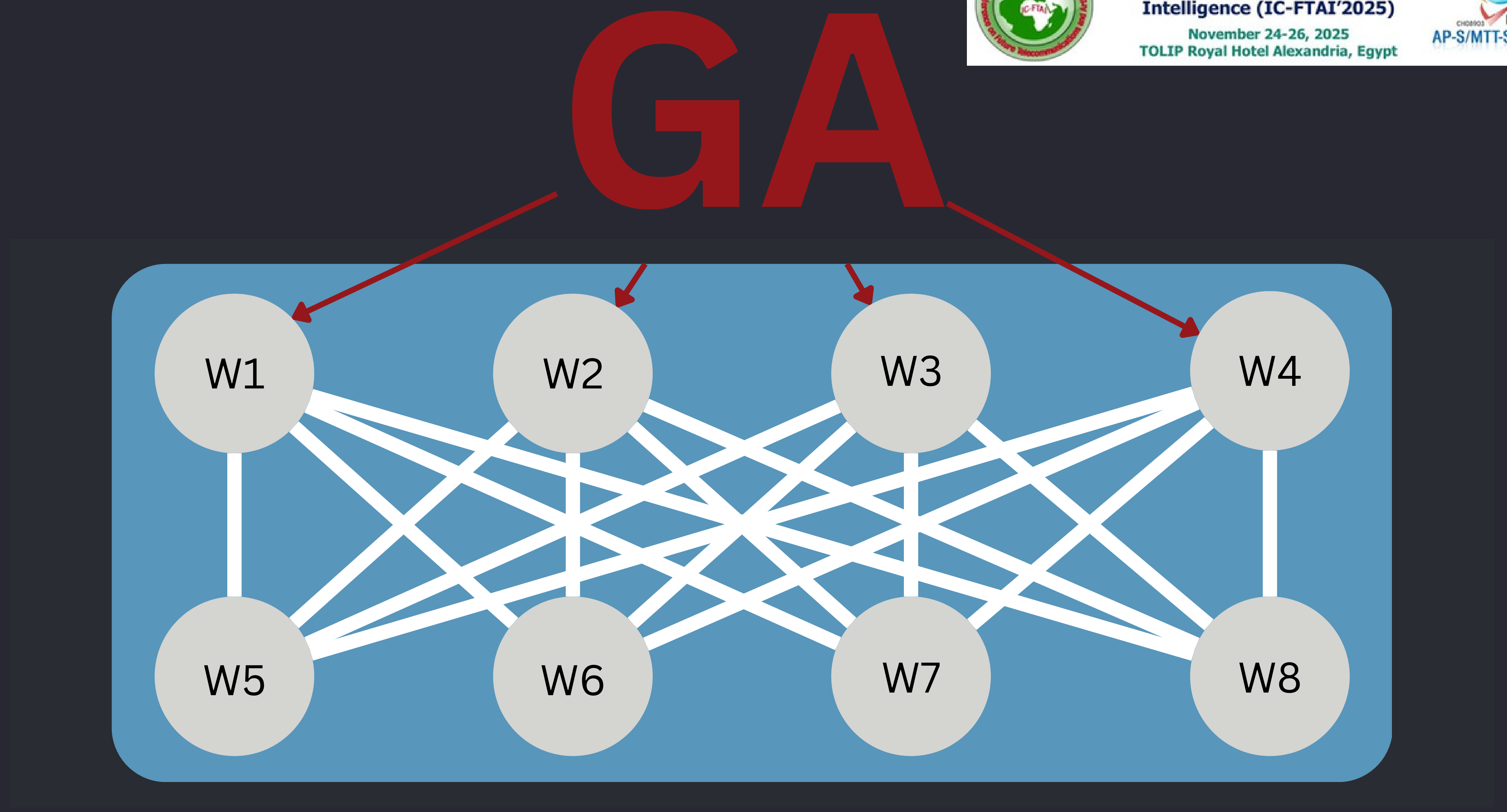

GA-Optimized VGG16 Technique: Genetic algorithms **optimize the weights of the model itself** across many evolutionary generations. **Each population** candidate represents a **distinct weight configuration** evaluated against validation accuracy. **Selection preserves elite performers**, **crossover generates offspring configurations**, and **mutation introduces exploration within the parameter space**.

Dataset & Preprocessing Pipeline

01

Data Source

Kaggle ChestX -ray Dataset: 5,857 images (4273 pneumonia-positive, 1584 normal)

02

Augmentation

Random rotation (±10°), brightness/contrast adjustment (±20%) and cropping to increase effective dataset size and model robustness.

03

Normalization

Pixel values scaled to [0,1] range; ImageNet statistics applied for transfer learning compatibility with pre-trained VGG16 weights.

04

Split Strategy

Training (70%),validation (15%), test (15%) stratified split ensuring balanced class representation across all subsets.

Hardware used

- GPU: RTX 2060
- Software: TensorFlow
- Batch size: 64
- Epochs: 10
- Population: 50 models
- Time for training: 16 hours

Genetic algorithm matched the metrics of other optimizers in only 3 generations!

Optimization Methods Comparison

Method	Speed(sec)	Accuracy	Robustness	Local Optima
Particle Swarm	24.04	Excellent	Moderate	Vulnerable
GA-PYGAD	24.55	Excellent	Very High	Avoids
GA-DEAP	23.84	High	Good	Avoids
Ant Colony	23.87	Excellent	High	Avoids
Bayesian	23.99	High	Good	Moderate
Grey Wolf	23.38	High	Moderate	Moderate

Excellent
Very High
High
Good
Moderate
Low

Genetic Algorithm selected for superior balance of convergence, robustness, and escape from local optima in high-dimensional CNN parameter spaces.

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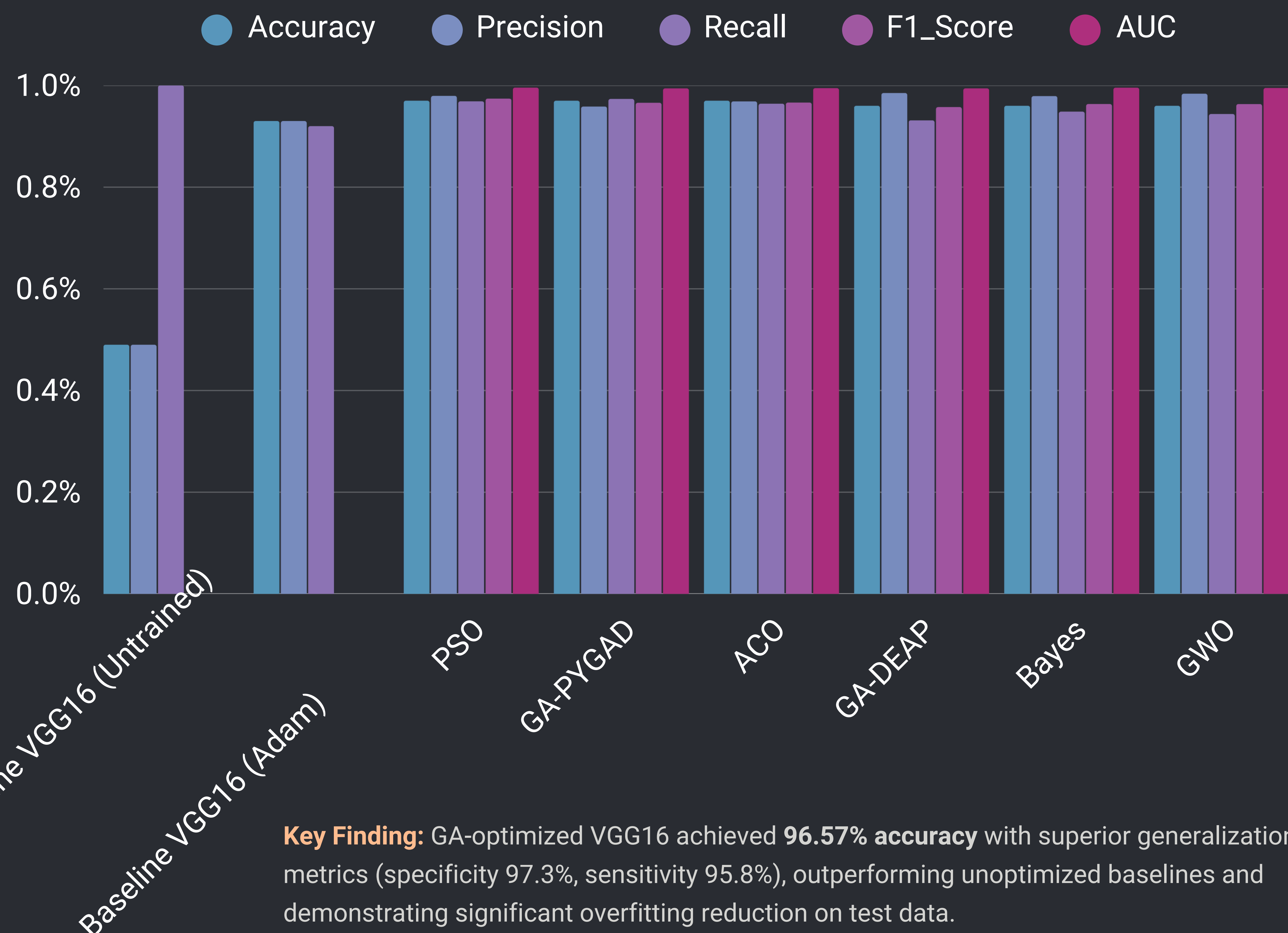
Experimental Results

Accuracy Comparison Across Optimization Methods:



Key Finding: GA-optimized VGG16 achieved **96.57% accuracy** with superior generalization metrics (specificity 97.3%, sensitivity 95.8%), outperforming unoptimized baselines and demonstrating significant overfitting reduction on test data.

✅ All Combined Results:

	Model	Accuracy	Precision	Recall	F1_Score	AUC	Inference_Time_ms	Size_MB
0	best_vgg16_aco_model	0.966459	0.968652	0.964119	0.966380	0.994685	23.080742	80.604563
1	best_vgg16_bayes_model	0.964119	0.979066	0.948518	0.963550	0.995607	23.914374	80.604562
2	best_vgg16_gwo_model	0.964119	0.983740	0.943838	0.963376	0.995035	22.480736	80.604564
3	best_vgg16_modelGA	0.965679	0.958525	0.973479	0.965944	0.994260	23.507520	80.604564
4	best_vgg16_model_deap	0.958658	0.985149	0.931357	0.957498	0.994268	23.285820	80.604565
5	best_vgg16_pso_model	0.974259	0.979495	0.968799	0.974118	0.995738	24.005320	80.604573

✅ Detailed results saved to 'model_results_detailed.csv'

--- Variance of Performance Metrics Across Models ---

Accuracy	2.565625e-05
Precision	1.048519e-04
Recall	2.681230e-04
F1_Score	2.936469e-05
AUC	4.139702e-07

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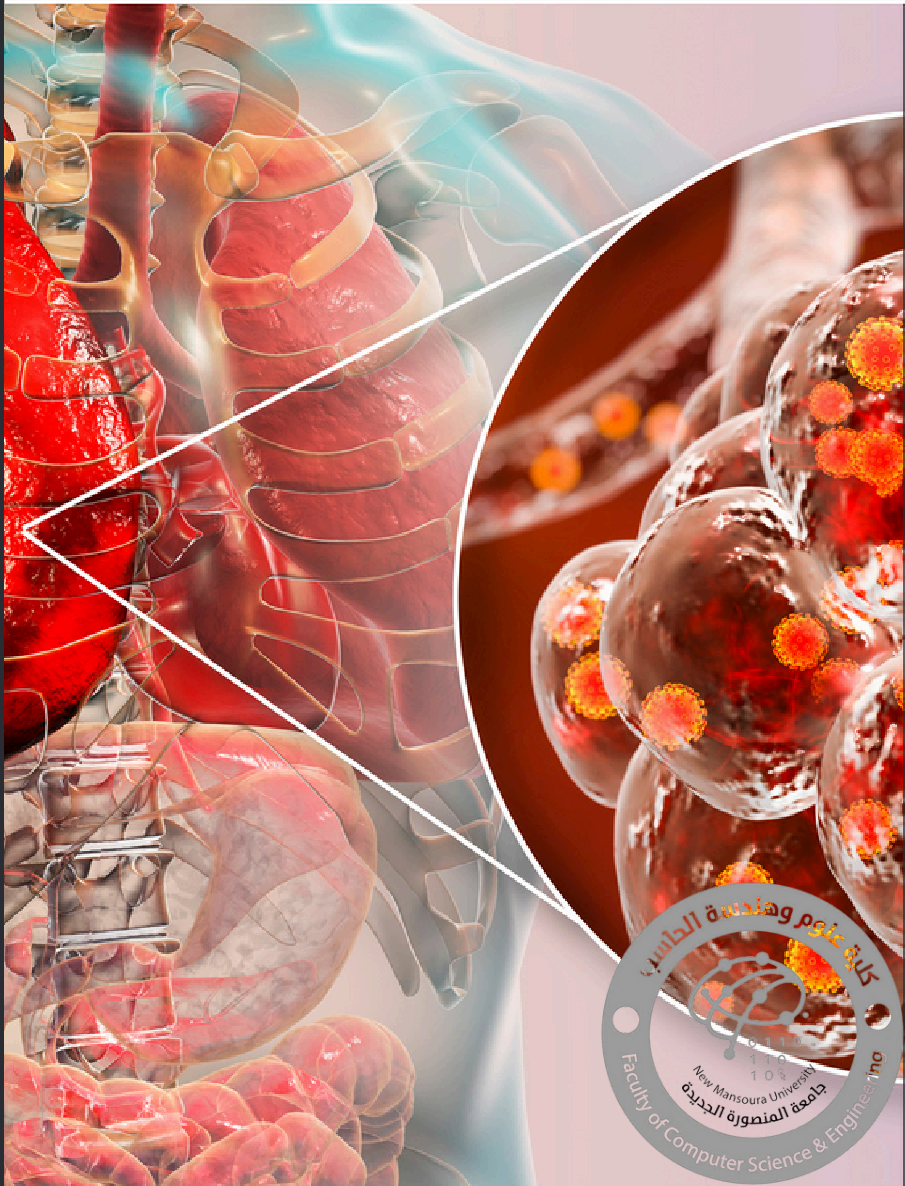
Performance Analysis & Clinical Insights

Strengths of GA Optimization

- Effective escape from local optima through population-based search
- Robust parameter configuration transferable across datasets
- Reduced overfitting: 3.36% improvement over unoptimized VGG16
- Balanced sensitivity-specificity trade-off for clinical deployment

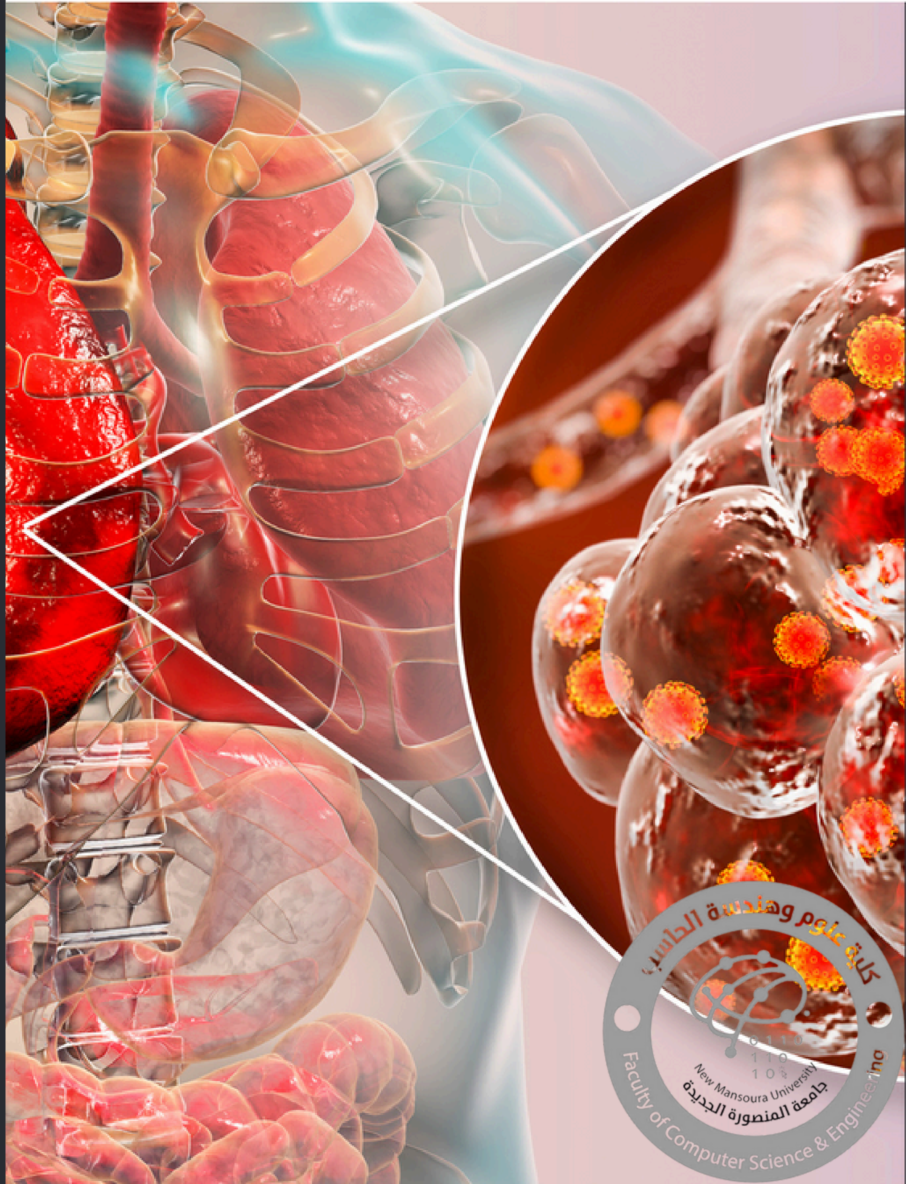
Practical Healthcare Implications

- Near-radiologist-level performance enabling decision support systems
- Reduced false positives minimizing unnecessary antibiotic prescriptions
- Enables rapid classification in resource-limited settings
- Foundation for real-time diagnostic assistants in clinical workflows



Ways to improve the model include:

- Using a bigger Model to capture more of the features in expense of longer training and classification time (real-time detection is not Mandatory)
- adding more data to reduce overfitting more and introduce other angles of x-ray like side views
- increasing the training time (hardware limitation on our end)
- Classify between types of pneumonia whether bacteria, virus, fungi or atypical (insufficient data)



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Conclusion & Future Directions

Key Findings

Genetic Algorithm optimization significantly improves CNN generalization for pneumonia detection, with strong clinical metrics and practical deployment readiness.

Hybrid Approaches

Future work will investigate ensemble methods combining GA with PSO, reinforcement learning for further accuracy gains, using up to date methods like Cross-validation for harsher testing and increasing compute to stress test overfitting.

Real-World Translation

Integration into hospital PACS systems and mobile diagnostic platforms for deployment in clinical and remote settings.

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Acknowledgments & Contact

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